

# ODE Test Problems Reference

Representative scalar, 2D, and 3D benchmark problems for testing ODE solvers. Solver recommendations are practical defaults, not exclusive choices.

## Scalar ODEs

| Problem                  | Type   | Equation(s), initial condition, and parameters  | Nonlinear | Exact | Stiff | Recommended solver(s)                                                                   | Purpose           |
|--------------------------|--------|-------------------------------------------------|-----------|-------|-------|-----------------------------------------------------------------------------------------|-------------------|
| Exponential growth/decay | Scalar | $u' = \lambda u$<br>$u(0)=1, \lambda=-2$        | No        | Yes   | No    | Any explicit RK: Euler for demos; RK4/Tsit5/Vern9 for convergence tests.                | Basic convergence |
| Linear forced equation   | Scalar | $u' = -u + \sin(t)$<br>$u(0)=0$                 | No        | Yes   | No    | Adaptive explicit RK, e.g. Tsit5/Vern9. Good for non-autonomous RHS tests.              | Forcing           |
| Logistic equation        | Scalar | $u' = r u(1-u/K)$<br>$u(0)=0.2, r=2, K=1$       | Yes       | Yes   | No    | Explicit RK methods; adaptive RK recommended near steep growth.                         | Bounded growth    |
| Riccati equation         | Scalar | $u' = u^2$<br>$u(0)=0.5$                        | Yes       | Yes   | No    | Adaptive explicit RK. Stop before blow-up time $t=2$ .                                  | Blow-up           |
| Bernoulli equation       | Scalar | $u' = -u + u^2$<br>$u(0)=0.5$                   | Yes       | Yes   | No    | Explicit RK. Useful for nonlinear exact-solution verification.                          | Nonlinear decay   |
| Gompertz equation        | Scalar | $u' = -r u \ln(u/K)$<br>$u(0)=0.2, r=1, K=1$    | Yes       | Yes   | No    | Explicit RK; ensure $u(t)>0$ to avoid log-domain issues.                                | Growth model      |
| Stiff linear equation    | Scalar | $u' = -1000(u - \cos(t)) - \sin(t)$<br>$u(0)=2$ | No        | Yes   | Yes   | Implicit Euler, Rosenbrock23/Rodas5, TRBDF2, BDF/QNDF. Explicit RK requires tiny steps. | Stiffness         |

## Two-Component ODE Systems

| Problem              | Type      | Equation(s), initial condition, and parameters                                                                        | Nonlinear | Exact | Stiff               | Recommended solver(s)                                                                      | Purpose         |
|----------------------|-----------|-----------------------------------------------------------------------------------------------------------------------|-----------|-------|---------------------|--------------------------------------------------------------------------------------------|-----------------|
| Harmonic oscillator  | 2D system | $x'=y$<br>$y'=-x$<br>$(x(0),y(0))=(1,0)$                                                                              | No        | Yes   | No                  | RK4/Tsit5 for accuracy; symplectic Euler/Störmer-Verlet for long-time energy behavior.     | Periodic motion |
| Damped oscillator    | 2D system | $x'=y$<br>$y'=-x-\gamma y$<br>$(x(0),y(0))=(1,0)$ , $\gamma=0.2$                                                      | No        | Yes   | No                  | Explicit RK; implicit methods also fine if damping is large.                               | Decay           |
| Linear saddle        | 2D system | $x'=x$<br>$y'=-y$<br>$(x(0),y(0))=(1,1)$                                                                              | No        | Yes   | No                  | Explicit RK. Use short intervals if growing component becomes large.                       | Stability       |
| Forced oscillator    | 2D system | $x'=y$<br>$y'=-x+\cos(t)$<br>$(x(0),y(0))=(0,0)$                                                                      | No        | Yes   | No                  | Adaptive explicit RK; good test for time-dependent RHS handling.                           | Forced motion   |
| Logistic + decay     | 2D system | $x'=2x(1-x)$<br>$y'=-y$<br>$(x(0),y(0))=(0.2,1)$                                                                      | Yes       | Yes   | No                  | Explicit RK. Good simple nonlinear system with exact solution.                             | Exact nonlinear |
| Riccati + decay      | 2D system | $x'=-x^2$<br>$y'=-y$<br>$(x(0),y(0))=(0.5,1)$                                                                         | Yes       | Yes   | No                  | Adaptive explicit RK. Stop before x blow-up time $t=2$ .                                   | Blow-up         |
| Triangular nonlinear | 2D system | $x'=-x$<br>$y'=-y+x^2$<br>$(x(0),y(0))=(1,0)$                                                                         | Yes       | Yes   | No                  | Explicit RK. Good for testing coupled exact solutions.                                     | Coupling        |
| Bernoulli coupled    | 2D system | $x'=-x$<br>$y'=-y+xy$<br>$(x(0),y(0))=(1,1)$                                                                          | Yes       | Yes   | No                  | Explicit RK. Tests multiplicative nonlinear coupling.                                      | Coupling        |
| Polar spiral         | 2D system | $r=\sqrt{x^2+y^2}$<br>$x'=-2y-rx$<br>$y'=2x-ry$<br>$(x(0),y(0))=(1,0)$                                                | Yes       | Yes   | No                  | Explicit RK. Good exact rotating nonlinear benchmark.                                      | Spiral          |
| Radial logistic      | 2D system | $r=\sqrt{x^2+y^2}$<br>$x'=x(1-r)$<br>$y'=y(1-r)$<br>$(x(0),y(0))=(0.5,0.5)$                                           | Yes       | Yes   | No                  | Explicit RK. Good for attracting invariant set tests.                                      | Unit circle     |
| Lotka-Volterra       | 2D system | $x'=\alpha x - \beta xy$<br>$y'=-\gamma y + \delta xy$<br>$(x(0),y(0))=(2,1)$<br>$\alpha=2$ , $\beta=\gamma=\delta=1$ | Yes       | No    | No                  | Explicit RK; high-order/adaptive RK for preserving phase accuracy. Monitor invariant.      | Predator-prey   |
| Van der Pol          | 2D system | $x'=y$<br>$y'=\mu(1-x^2)y-x$<br>$(x(0),y(0))=(2,0)$<br>$\mu=1$ or $100$                                               | Yes       | No    | Yes for large $\mu$ | $\mu=1$ : explicit RK. $\mu=100$ : Rosenbrock/Rodas5, TRBDF2, BDF/QNDF.                    | Limit cycle     |
| Brusselator          | 2D system | $x'=A-(B+1)x+x^2y$<br>$y'=Bx-x^2y$<br>$(x(0),y(0))=(1.2,3.1)$<br>$A=1$ , $B=3$                                        | Yes       | No    | Usually no          | Adaptive explicit RK for nonstiff regimes; Rosenbrock/BDF if parameters make it stiff.     | Chemical osc.   |
| FitzHugh-Nagumo      | 2D system | $v'=v-v^3/3-w+l$<br>$w'=\epsilon(v+a-bw)$<br>$(v(0),w(0))=(0,0)$<br>$\epsilon=0.08$ , $a=0.7$ , $b=0.8$ , $l=0.5$     | Yes       | No    | Moderate            | Adaptive RK for moderate $\epsilon$ ; Rosenbrock/BDF for small $\epsilon$ or sharp spikes. | Excitability    |
| Nonlinear pendulum   | 2D system | $\theta'=\omega$<br>$\omega'=-\sin(\theta)$<br>$(\theta(0),\omega(0))=(\pi/2,0)$                                      | Yes       | No    | No                  | RK4/Tsit5 for short runs; symplectic methods for long-time energy behavior.                | Hamiltonian     |
| Duffing oscillator   | 2D system | $x'=y$<br>$y'=x-x^3-\delta y$<br>$(x(0),y(0))=(1,0)$ , $\delta=0.2$                                                   | Yes       | No    | No                  | Adaptive explicit RK. For forced/chaotic variants use tight tolerances.                    | Bistability     |

## Three-Component ODE Systems

| Problem            | Type      | Equation(s), initial condition, and parameters                                                                                                                                   | Nonlinear | Exact | Stiff        | Recommended solver(s)                                                                                             | Purpose          |
|--------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|--------------|-------------------------------------------------------------------------------------------------------------------|------------------|
| Lorenz system      | 3D system | $x' = \sigma(y-x)$<br>$y' = x(\rho-z)-y$<br>$z' = xy-\beta z$<br>$(x(0), y(0), z(0)) = (1, 1, 1)$<br>$\sigma=10, \rho=28, \beta=8/3$                                             | Yes       | No    | No           | Adaptive explicit RK with tight tolerances, e.g. Vern9/Tsit5. Chaotic: compare statistically or over short times. | Chaos            |
| Rössler system     | 3D system | $x' = -y-z$<br>$y' = x+a y$<br>$z' = b+z(x-c)$<br>$(x(0), y(0), z(0)) = (1, 0, 0)$<br>$a=0.2, b=0.2, c=5.7$                                                                      | Yes       | No    | No           | Adaptive explicit RK. Use tight tolerances for chaotic trajectories.                                              | Chaos            |
| SIR epidemic model | 3D system | $S' = -\beta S I$<br>$I' = \beta S I - \gamma I$<br>$R' = \gamma I$<br>$(S(0), I(0), R(0)) = (0.99, 0.01, 0)$<br>$\beta=3, \gamma=1$                                             | Yes       | No    | No           | Explicit RK. Check conservation $S+I+R=1$ and positivity.                                                         | Conservation     |
| Robertson kinetics | 3D system | $y_1' = -0.04y_1 + 1e4 y_2 y_3$<br>$y_2' = 0.04y_1 - 1e4 y_2 y_3 - 3e7 y_2^2$<br>$y_3' = 3e7 y_2^2$<br>$(y_1(0), y_2(0), y_3(0)) = (1, 0, 0)$                                    | Yes       | No    | Yes          | Classic stiff test: Rosenbrock23/Rodas5, TRBDF2, BDF/QNDF. Avoid explicit RK except for demos.                    | Severe stiffness |
| Oregonator         | 3D system | $x' = s(y+xy-qx^2)$<br>$y' = (-y-xy+fz)/s$<br>$z' = w(x-z)$<br>$(x(0), y(0), z(0)) = (1, 2, 3)$<br>$s=77.27, w=0.161, q=8.375e-6, f=1$                                           | Yes       | No    | Yes/Moderate | Rosenbrock/Rodas5 or BDF for robust integration; adaptive RK may work with small steps.                           | Chemical osc.    |
| Hindmarsh-Rose     | 3D system | $x' = y - a x^3 + b x^2 - z + I$<br>$y' = c - dx^2 - y$<br>$z' = r(s(x-xR) - z)$<br>$(x(0), y(0), z(0)) = (-1.6, -10, 1)$<br>$a=1, b=3, c=1, d=5, r=0.006, s=4, xR=-1.6, I=3.25$ | Yes       | No    | Multi-scale  | Adaptive RK for qualitative runs; Rosenbrock/BDF if slow-fast stiffness causes difficulty.                        | Bursting         |

Notes: For chaotic systems, pointwise agreement over long time intervals is not a reliable pass/fail test; compare short-time accuracy, invariants where available, or statistical/qualitative features. For stiff problems, implicit or linearly implicit methods are generally preferred.